

Southern Ontario city dispatch

A dock delay is two deadlines, not one.

A detention & hours-of-service exception desk. The billing clock and the driver's legal clock, on one time axis — because the operational deadline arrives first.



The dock costs a city desk twice

and only one of them shows up in the accounting

The visible loss

Detention that is never billed

The truck waits past the two-hour free allowance. Nobody wrote down the in and out times, so there is no defensible claim — and the carrier eats it. Every TMS on the market can compute this after someone types the timestamps in. Nobody types them in.

The invisible loss

The next load

A driver can have driving hours left while the dock wait eats the 14-hour on-duty window. The next pickup becomes infeasible before the free time even runs out — so the desk finds out when the driver calls, hours after it could have been fixed.

The brief calls automated detention billing the critical requirement. We took the bonus question seriously instead: **what happens when a driver runs out of hours sitting in a dock queue?**

Replayed against Road Star's own export

3,938 orders · 10,397 legs · 131 drivers · 2026-07-03 → 08-28

Charges drafted, per 30 days

\$31k

257 charges over 56 days, \$57,638 total. 768.5 billable hours at \$75/h, floored to 15-minute increments.

Warned before billing started

115 of 126

Stops that went past free time, flagged 30 minutes early. Recall 91%, precision 79% — trained on the first four weeks, scored on the last four.

Legs that ran empty

35.5%

6,756 loaded against 3,723 empty. The relief driver we offer is often already deadheading past the dock.

These are not our estimates. Every figure is produced by the code in the repo, running the organizers' file through the rules. The one number we will not say is **"unbilled"** — the export has no billing records, so we say **exposure**.

The money is in a thin tail, and the tail is bursty

which is exactly what manual logging misses

Median dwell	most stops are nowhere near the line	< 1h
Delivery p90	just past the two-hour allowance	2.03h
July, drafted	busiest week alone: \$15,938 · 54 stops over	\$41.9k
August, drafted	same fleet, same customers	\$15.0k

July drafted **nearly three times** what August did. A desk that watches averages plans for the wrong month — and a desk that watches nothing bills neither.



Every stop in the export, replayed through the rules — in the product, on the Data page.

One truck, one day, both clocks

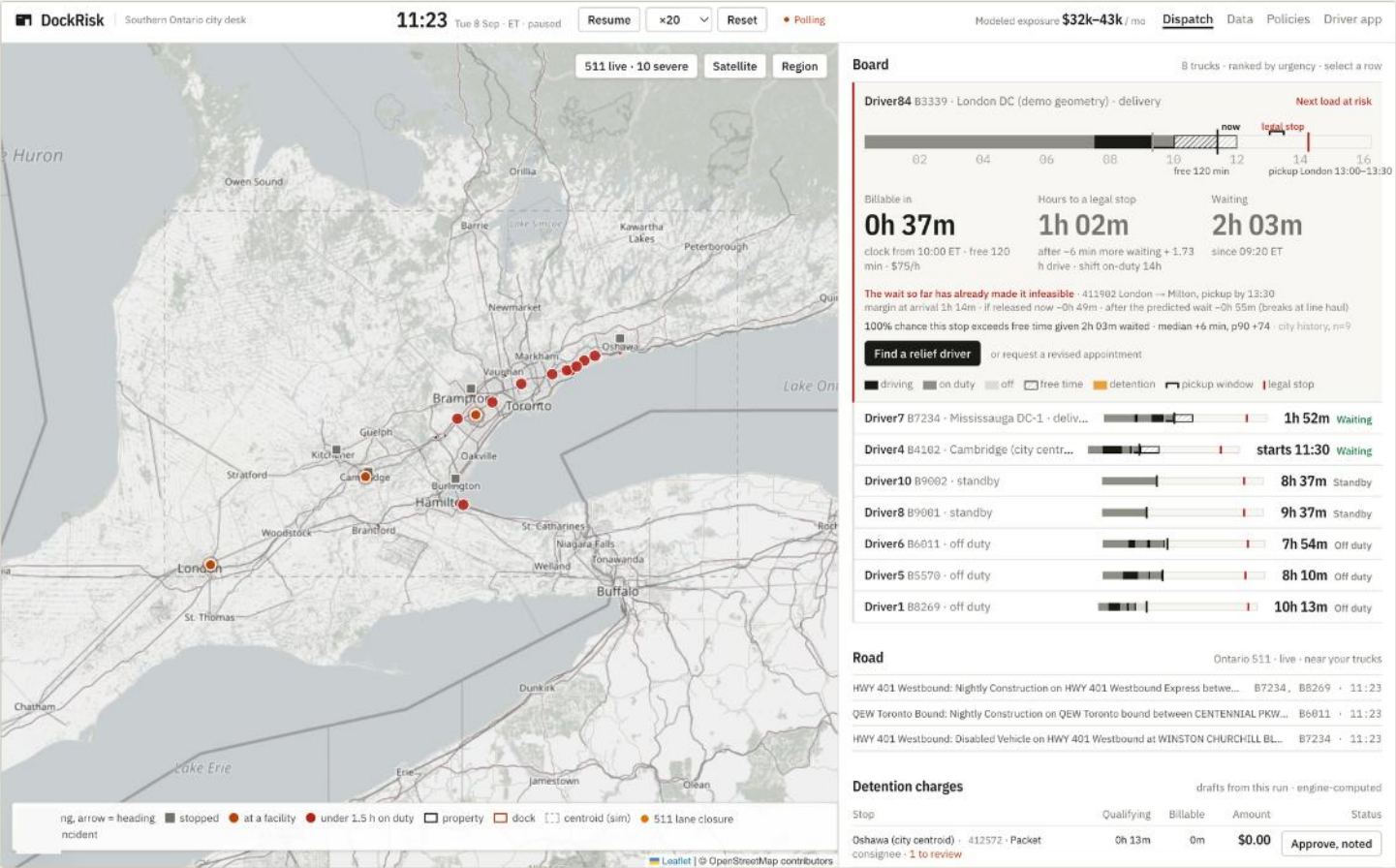
The operational deadline arrives first.



37 minutes. That is the gap in the demo scenario — the next load dies at 12:53, the free time does not run out until 13:30. Every detention product on the market starts paying attention at 13:30. By then the load is gone and the only question left is who eats it.

One board, ranked by urgency

every truck with telemetry is one row — the alert is the row, not a separate list



Columns are the deadlines first — billable in · hours to a legal stop · waiting. The binding one is ink, the other is grey. Physical dwell comes last, because it is context.

Driver84 · B3339 · London DC · delivery

11:23 ET, in the scenario

Billable in	Hours to a legal stop	Waiting
0h 37m	1h 02m	2h 03m
clock from 10:00 ET · free 120 min · \$75/h	after ~6 min more waiting + 1.73 h drive · shift on-duty 14h	since 09:20 ET

The wait so far has already made it infeasible

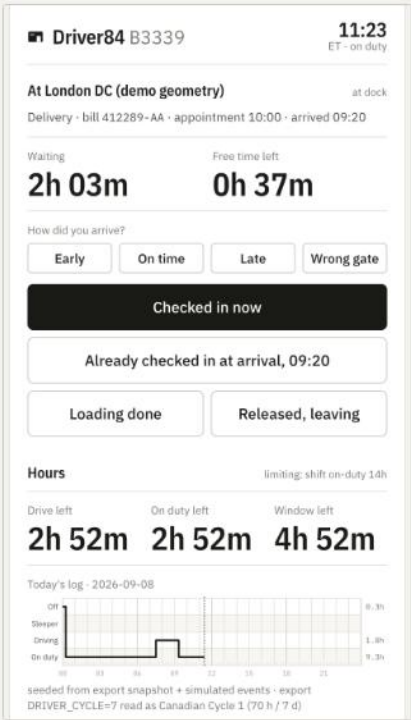
411902 London → Milton, pickup by 13:30 · margin at arrival **1h 14m** · if released now **-0h 49m** · after the predicted wait **-0h 55m**

100% chance this stop exceeds free time given 2h 03m waited · median +6 min, p90 +74 · city history, n=9

Three margins, not one verdict: **at arrival** tells the dispatcher this was always tight; **if released now** is the best case; **after the predicted wait** is the honest one. All three are shown because a dispatcher who is about to move a load wants to see the argument, not the answer.

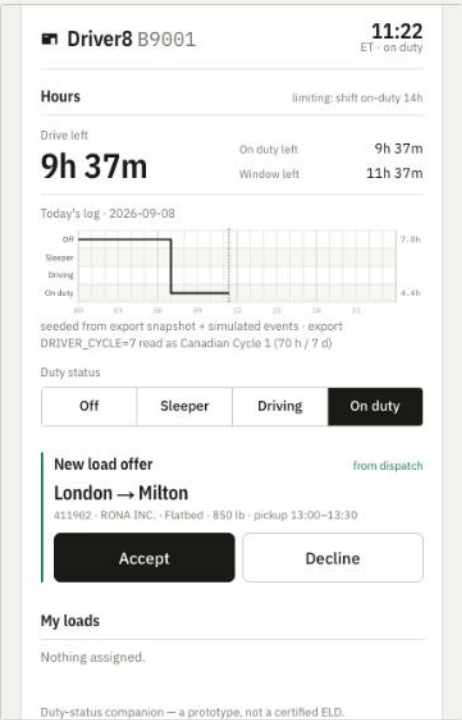
The same two clocks, in the cab

the driver interface the brief requires — ELD-style log, duty status, load acceptance



Arrival class, check-in, door, done, released — four taps, and each one is a source in the evidence ledger.

Duty-status companion — a prototype, not a certified ELD. Labelled that way in the product too.

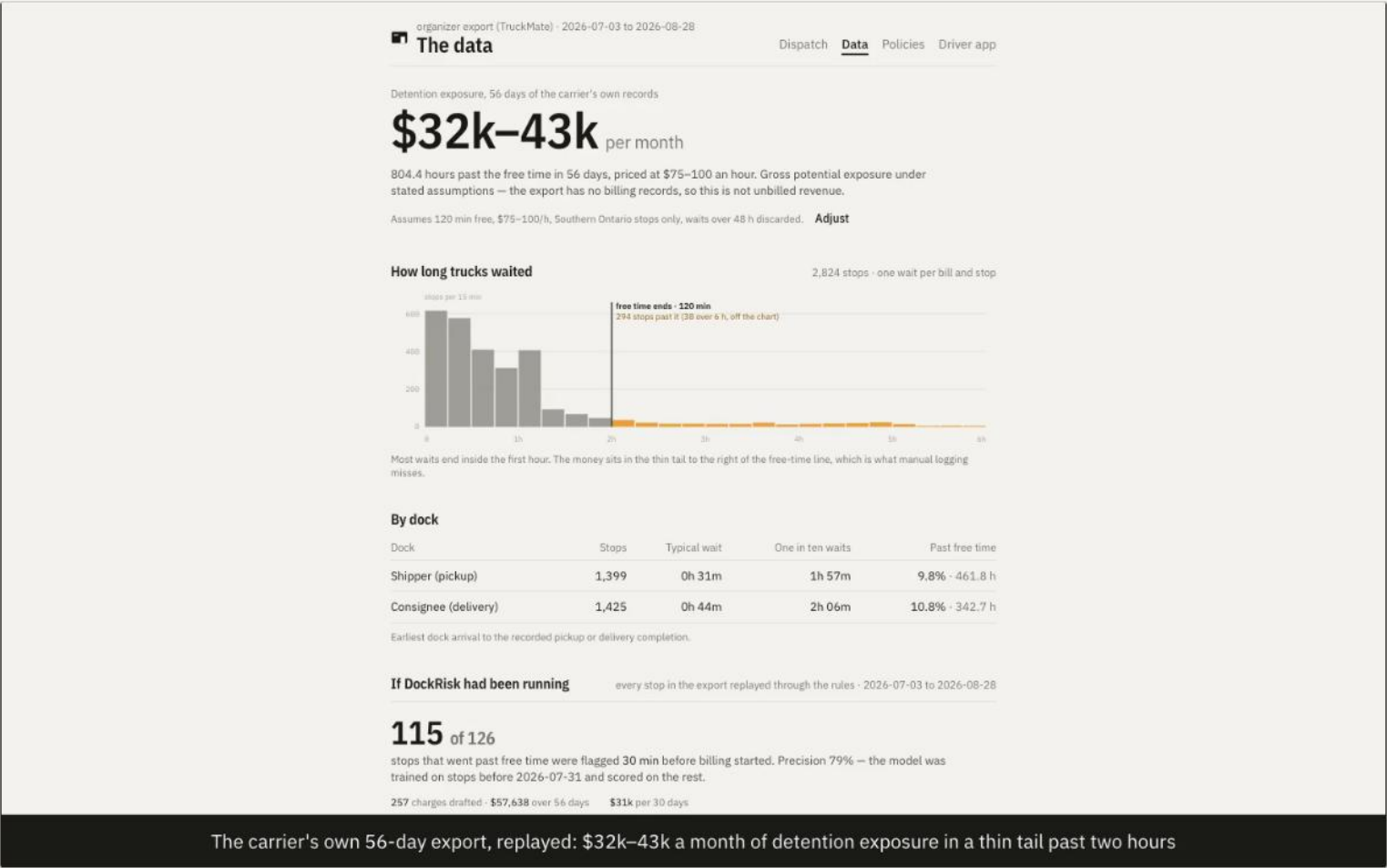


The relief driver gets one offer, with the load on it. Dispatch ranked the candidates; the driver still decides.

Switch to the app

Live demo

Arrival → driver check-in → the two clocks collide with a live 401 event → offer and relief → release and charge → the evidence packet. One scenario, start to finish, on the simulator's clock.



If the live app stalls: press play here. 90 seconds, same scenario, recorded. Do not apologise — just keep talking over it.

A charge is only worth what its evidence is worth

the packet reads as an invoice, not a rules dump

Physical dwell	10:07 → 11:14	1h 07m
Qualifying dwell	clock from the later of check-in and appointment	0h 13m
Free time	15-min increments, floored · on-time arrival required	2h 00m

Needs review before billing
facility geometry is centroid (confidence 0.3), not verified dock geometry

Every line carries its rule in words. Below this sits the event ledger — every transition, who reported it (GPS geofence, driver app, engine) and when it was received.

DockRisk · detention claim · draft · 2026-09-08

Dispatch

Print / PDF

Oshawa (city centroid)

Delivery · bill 412572 · Driver5 · B5570 · visit #190

\$0.00

0 billable minutes at \$75/h

in 10:07

appointment 11:00

release 11:14

10:00

10:15

10:30

10:45

11:00

11:15

11:30

free 120 min from 11:00

The visit on a time axis: hatched is free time from the clock start, amber is billable. Ticks are the events in the timeline below.

Physical dwell

1h 07m

10:07 → 11:14

Qualifying dwell

0h 13m

clock from the later of check-in and appointment (11:00) to release (11:12)

Free time

2h 00m

15-min increments, rounded floor · on-time arrival required · policy: default

Needs review before billing

facility geometry is centroid (confidence 0.3), not verified dock geometry

Evidence confidence 1 · facility outline: centroid (not surveyed)

Timeline

The amount first, then the arithmetic, then the proof.

Two AI moments. The engine computes every number.

both labelled in the UI · both with a deterministic fallback

1 · Read the rate confirmation

A carrier's rate can become structured detention terms — free time, rate, increment, clock-start rule — with the **clause quoted under each field**, so the dispatcher confirms against the contract, not against the model.

2 · Draft the customer notice

Written from the evidence packet that already exists. Every figure in the draft is checked against the engine's own output before it is shown.

The rule we did not break: the model never produces an amount, a duration or a deadline. When the endpoint is absent the rules path runs and the product keeps working.

Notice to the customer

drafted by gpt-6-astra · edit before sending

Detention notice — Bill 412289-AA — London DC (demo geometry)

Hello,

For bill 412289-AA at London DC (demo geometry) on 2026-09-08, unit B3339 recorded arrival at 09:25 and departure at 12:16. Please confirm these in/out times and provide confirmation of the driver's check-in time.

Subject to confirmation, the detention calculation uses a clock start of 10:00 and clock end of 12:16, with qualifying dwell of 136.0 minutes. The default policy provides 120 minutes of free time at a rate of 75.0 per hour in 15-minute increments. The calculated billable time is 15 minutes, for an amount of 18.75.

The detention amount remains subject to confirmation of driver check-in and on-time status. Thank you for reviewing.

Before sending, confirm: on-time status not confirmed by driver; no check-in confirmation; clock started at property entry; required evidence missing: driver check-in; Confirm that the default policy applies before sending.; Resolve the discrepancy between the review reason stating that the clock started at property entry and the supplied clock_start of 10:00; the recorded arrival is 09:25..

Copy

Draft again

Marked as drafted, with the confirm-before-sending list underneath — including the discrepancies the engine found in its own inputs.

Detention terms are per customer, so they are data — not code

free time · rate · increment · clock-start rule · on-time requirement

Every carrier we read about argues the same four things with every shipper: when the clock starts, how long is free, what it costs, and whether the driver has to have been on time. So those are fields, editable at the desk, versioned, and **replay-safe** — change the policy and the historical replay recomputes rather than rewriting the past.

Of the 257 charges drafted from the export, **225 had an appointment on file** and would start the clock at the appointment; **32 would start it at arrival**. Same fleet, same month, two different rules — which is exactly why this cannot be hard-coded.

Policies

Detention terms

DispatchDataPoliciesDriver app

The model reads a rate confirmation and drafts the detention terms with the clause it relied on under each field. You confirm them. The engine, not the model, computes every charge.

Customer

ACME Foods

Extracted terms appear here for review.

Rate confirmation or agreement text

RATE CONFIRMATION — Lane: Milton, ON to London, ON.
Rate: \$650 flat.
Detention: 2 hours free time at shipper and consignee. \$85.00 per hour thereafter, billed in 30-minute increments, not to exceed \$500 per stop. Detention counts from the later of driver check-in and the scheduled appointment time. Driver must be on time for the appointment; detention is not payable if the driver is late.
Carrier must provide in/out times signed by the facility and the BOL with the invoice.

Extract terms

Active policies

facility overrides customer overrides default

Scope	Free	Rate	Incr.	Cap	Clock starts at	Source
Default	120 min	\$75/h	15 min	—	later of check-in and appointment, on-time required	default
ACME Foods	120 min	\$85/h	30 min	\$500	later of check-in and appointment, on-time required	confirmed · dispatcher (from extracted-rules)

Change a term, and every draft charge behind it recomputes.

What we did not build, and why

named in the product, not just in this deck

Axle-weight compliance

131 of 131 rows

The Trucks sheet has exactly one column — a truck number. There are no specs in the file, so the check is not buildable. Trailer-capacity checks are, and those we do.

Leg-level hours of service

100%

Our first "finding" was that the export's leg HOS was dangerously stale. It failed its own test: it is a frozen per-driver copy, not a decision field. We kept the retraction in the repo and join hours from the driver sheet instead.

Charges against a city centroid

to review

Where we only have a city point rather than surveyed dock geometry, the charge is drafted but flagged **needs review** and never billed silently. s.76 adverse-driving relief is surfaced as "possible, review required" — never applied automatically.

No rates or revenue exist anywhere in the export, and 68% of legs carry a 1980 sentinel instead of a planned date. Financial impact is **modelled from dwell and distance, and labelled as modelled**, everywhere it appears.

Every required component, and the bonus

Hackathon_Project_Brief.pdf · City Dispatch Workflow & Fleet Automation

1 · Web dispatcher dashboard	Leaflet map with the satellite toggle , breadcrumbs, per-leg distance and speed, geofences, automated detention past the 2-hour FTL allowance	built
2 · Separated simulation engine	its own service, one virtual clock, OSRM road geometry, scripted dock dwell and 401 slowdown, deterministic replay from a seed	built
3 · Driver interface	responsive companion: ELD-style log grid, duty status, load specs, acceptance	built
4 · Pipeline & integration	TruckMate workbook import, Ontario 511 live events, OSRM, Nominatim — consolidated on one board	built
Bonus · edge-case discovery	"what happens when a driver runs out of HOS while sitting in a dock queue?" and "unexpected 401 closures" — both are the product, not a footnote	the thesis

Canadian HOS recomputed from duty events under SOR/2005-313 — 13/14/16 h, 10 h off with an 8 h core, Cycle 1 and 2. Used **forward**, not just for compliance reporting.

Three services, one clock

everything below runs on a laptop with no network, except the live road feed

Engine	Python 3.12 · FastAPI · SQLite · Shapely for the geofences · pandas/openpyxl for the TruckMate workbook
Web	Next.js 16 · React 19 · Tailwind 4 · Leaflet on OpenStreetMap and Esri imagery · IBM Plex
Simulator	a separate service, as the brief requires — one virtual clock, deterministic from a seed, survives an API restart
Outside	OSRM road geometry · Ontario 511 live events · Nominatim · an OpenAI-compatible endpoint for the two language tasks
Deployed	Docker on a VPS behind a Cloudflare Tunnel for the API; Vercel for the web

The state machine, not a timestamp diff

approach → entered → checked in → at dock → done → released → exited. Debounced against GPS jitter, so a truck circling a yard does not open and close three visits.

One dwell per bill and stop

Not per leg. Multi-leg orders otherwise inflate the charge count roughly four-fold — a bug we shipped, found in replay, and fixed.

To close

Watch the clock that runs out first.

Detention timers are table stakes. Connecting the dock clock to the next dispatch decision — **in time to act on it** — is the product. On Road Star's own 56 days that is \$31k a month of exposure and 115 of 126 stops caught before the billing even started.



Junsu Park · Braemar College

RoadStar Hackathon 2026 · Spur Innovation Center, Waterloo

Live dockrisk.vercel.app

Code github.com/Ketchio-dev/dockrisk

Data the public demo runs on a synthetic sample of the same shape — the organizers' export stays private